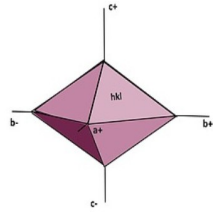
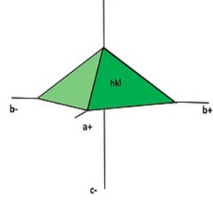
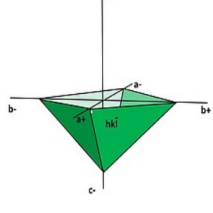
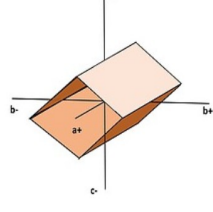
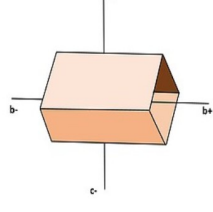
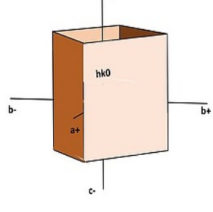
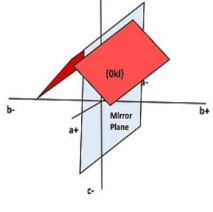
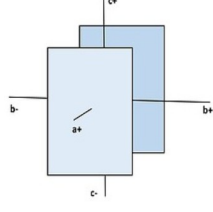
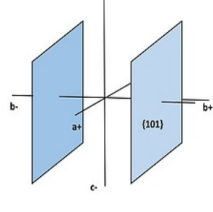
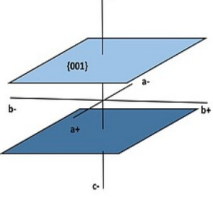
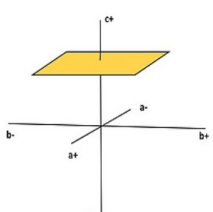


Systèmes Cristallins: Exemples de Géométrie

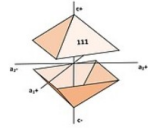
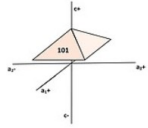
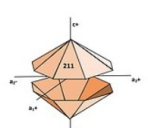
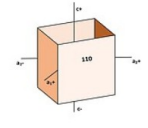
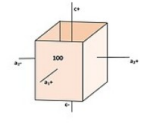
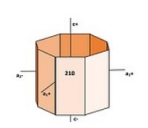
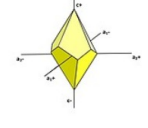
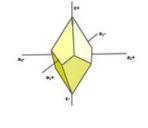
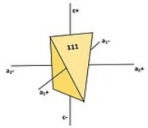
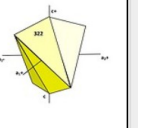
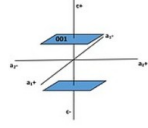
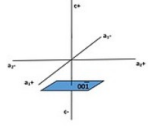
Système Orthorhombique ($\alpha = \beta = \gamma = 90^\circ$)

Formes fréquentes

Dipyramid & Pyramids	 Orthorhombic dipyramid	 Orthorhombic Pyramid {hkl}; Positive	 Orthorhombic Pyramid {hkl}; Negative	
Prisms & Domes	 Orthorhombic Prism {0kl}; 1st Order	 Orthorhombic Prism {h0l}; 2nd Order	 Orthorhombic Prism {hk0}; 3rd Order	 Orthorhombic Dome {0kl}; 1st Order
Pinacoids & Pedions	 Orthorhombic Pinacoid {100}	 Orthorhombic Pinacoid {010}	 Orthorhombic Pinacoid {001}	 Pedion

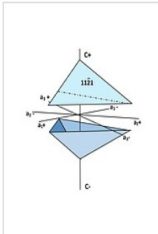
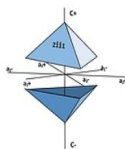
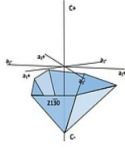
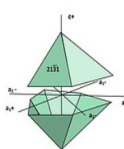
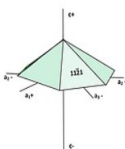
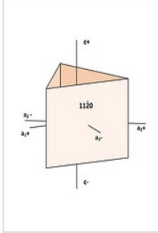
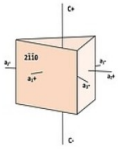
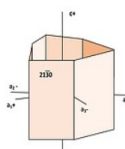
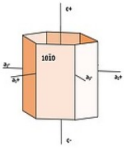
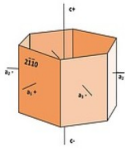
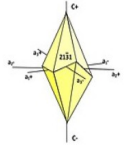
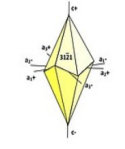
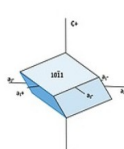
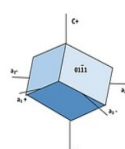
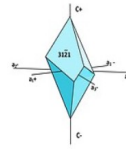
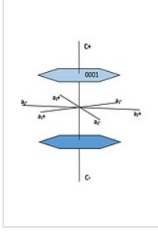
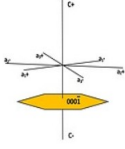
Système Tétragonal ($a = b$, $\alpha = \beta = \gamma = 90^\circ$)

Formes fréquentes

Pyramids & Dipyramids				
	Tetragonal Dipyramid 1st Order	Tetragonal Pyramid 2nd Order Positive	Ditetragonal Dipyramid	
Prisms & Diprisms				
	Tetragonal Prism 1st Order	Tetragonal Prism 2nd Order	Ditetragonal Prism	
Trapezohedrons; Disphenoid; & Scalenohedron				
	Tetragonal Trapezohedron Left	Tetragonal Trapezohedron, Right	Tetragonal Disphenoid	Tetragonal Scalenohedron Positive
Pinacoid & Pedion				
	Tetragonal Pinacoid	Tetragonal Pedion, basal		

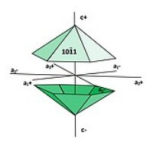
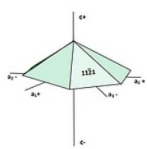
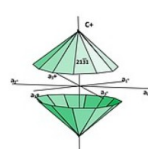
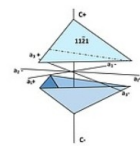
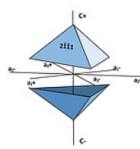
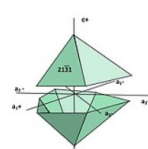
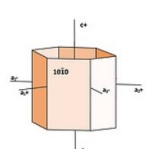
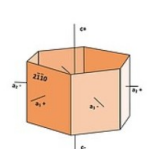
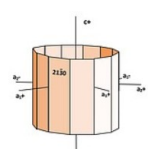
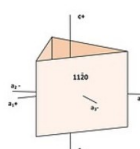
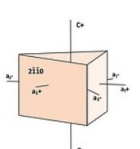
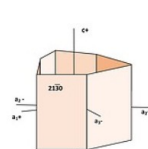
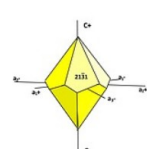
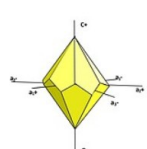
Système Trigonal ($a = b$, $\alpha = \beta = \gamma = 90^\circ$)

Formes fréquentes

Pyramids and Dipyramids	 Trigonal Dipyramid: 1st Order	 Trigonal Dipyramid 2nd Order	 Ditrigonal Pyramid 1st Order	 Ditrigonal Dipyramid	 Hexagonal Pyramid: 2nd Order		
Prisms & Diprisms	 Trigonal Prism: 1st Order	 Trigonal Prism: 2nd Order	 Ditrigonal Prism	 Hexagonal Prism 1st Order	 Hexagonal Prism 2nd Order		
Scalenohedrons	Rhombohedrons & Trapezohedrons		 Trigonal Scalenohedron Positive	 Trigonal Scalenohedron Negative	 Rhombohedron: Positive	 Rhombohedron: Negative	 Trigonal Trapezohedron: Left Positive
Pinacoids & Pedions	 Hexagonal Pinacoid (001)	 Hexagonal Pedion					

Système Hexagonal ($a = b$, $\alpha = \beta = 90^\circ$, $\gamma = 120^\circ$)

Formes fréquentes

Pyramids & Dipyramids						
	Hexagonal Dipyramid 1st Order	Hexagonal Pyramid 2nd Order	Dihexagonal Dipyramid	Trigonal Dipyramid: 1st Order	Trigonal Dipyramid 2nd Order	Ditrigonal Dipyramid
Prisms & Diprisms						
	Hexagonal Prism 1st Order	Hexagonal Prism 2nd Order	Dihexagonal Diprism 1st Order	Trigonal Prism: 1st Order	Trigonal Prism: 2nd Order	Ditrigonal Prism
Trapezohedron						
	Hexagonal Trapezohedron Left	Hexagonal Trapezohedron Right				
Pinacoid & Pedion						
	Hexagonal Pinacoid (001)	Hexagonal Pedion				

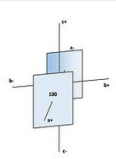
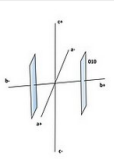
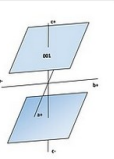
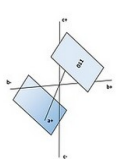
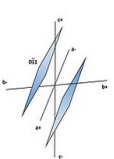
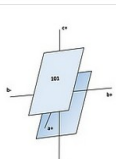
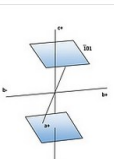
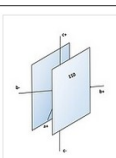
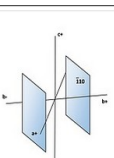
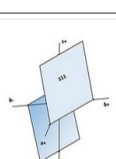
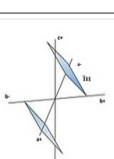
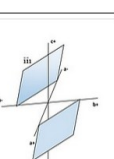
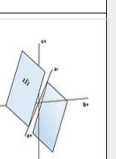
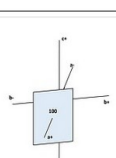
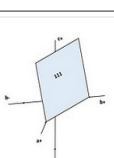
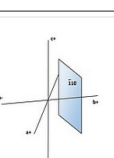
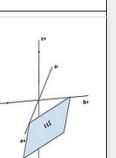
Système Monoclinique ($\alpha = \gamma = 90^\circ$)

Formes fréquentes

Prisms & Domes				
	{hkl}; 4th Order Prism Positive	{0kl}; 1st Order Prism	{hk0}; 3rd Order Prism	3rd Order Dome
Pinacoids				
	{h0l}; 2nd Order Pinacoid Positive	{100}; a Pinacoid	{010}; b Pinacoid	{001}; c Pinacoid; Basal Pinacoid

Système Triclinique (-)

Formes fréquentes

Triclinic Pinacoid {h00} {0k0} {00l}; (3 of 3)				
	Front {100}	Side {010}	Basal {001}	
Triclinic Pinacoid {0kl} 1st Order; (2 of 2)				
	Positive {011}	Negative {010}		
Triclinic Pinacoid {h0l} 2nd Order; (2 of 2)				
	Positive {101}	Negative {101}		
Triclinic Pinacoid {hk0} 3rd Order; (2 of 2)				
	Positive {110}	Negative {110}		
Triclinic Pinacoid {hkl} 4th Order; (4 of 4)				
	Positive Right {111}	Negative Right {111}	Negative Left {111}	Positive Left {111}
Triclinic Pedions; (4 of 26)				
	{100}	{111}	{110}	{111}

Système Isométrique ($a = b = c$, $\alpha = \beta = \gamma = 90^\circ$)

Formes fréquentes

Hexoctahedral Class							
	Isometric Hexahedron	Isometric Octahedron	Isometric Dodecahedron	Isometric Tetrahexahedron	Isometric Trisoctahedron	Isometric Trapezohedron	Isometric Hexoctahedron
Hextetrahedral Class							
	Isometric Tetrahedron: Positive	Isometric Tetrahedron: Negative	Isometric Tristetrahedron: Positive	Isometric Deltoid Dodecahedron: Positive	Isometric Hextetrahedron: Positive		
Diploidal Class							
	Isometric Pyritohedron Positive	Isometric Striated Pyrite	Isometric Diploid				
Tetartoidal Class							
	Isometric Tetartohedron Right Positive						

Groupes d'Espace, Systèmes Cristallins et Réseaux de Bravais

Famille Cristalline	Système Cristallin	Système de Réseau	Groupes d'Espace	Réseaux de Bravais	Notation Hermann-Mauguin
Isométrique	Isométrique	Isométrique	36	P, I, F	m-3m, 4/m-3-2/m, 432, -43m, 23
Tétragonal	Tétragonal	Tétragonal	68	P, I	4/mmm, 422, 4mm, -42m, 42m, 4/m, 4
Hexagonal	Hexagonal	Hexagonal	27	P	6/mmm, 622, 6mm, -6m2, 6/m, -6, 6
Hexagonal	Trigonal	Hexagonal	25	P	-3m, 3m, 32, -3, 3
Hexagonal	Trigonal	Rhomboédrique	7	R	-3m, 3m, 32, -3, 3
Orthorhombique	Orthorhombique	Orthorhombique	59	P, C, I, F	mmm, 222, 2mm
Monoclinique	Monoclinique	Monoclinique	13	P, C	2/m, m, 2
Triclinique	Triclinique	Triclinique	2	P	-1, 1

Les groupes d'espace en trois dimensions sont formés par des combinaisons des 32 groupes ponctuels cristallographiques avec les 14 réseaux de Bravais, chacun de ces derniers appartenant à l'un des 7 systèmes de réseau. Cela signifie que l'action de tout élément d'un groupe d'espace donné peut être exprimée comme l'action d'un élément du groupe ponctuel approprié, suivie éventuellement d'une translation.

Un groupe d'espace est donc une combinaison de la symétrie translationnelle d'une maille élémentaire (y compris le centrage du réseau), des opérations de symétrie du groupe ponctuel (réflexion, rotation et rotation impropre, également appelée roto-inversion), ainsi que des opérations de symétrie d'axe hélicoïdal et de plan de glissement.

Les groupes d'espace (groupes de symétrie d'une configuration dans l'espace) sont classés en systèmes cristallins selon leurs groupes ponctuels, et en systèmes de réseau selon leurs réseaux de Bravais. Les systèmes cristallins qui ont des groupes d'espace attribués à un système de réseau commun sont combinés en une famille cristalline.

Classes Cristallines, Symétries et Intuitions

Système Cristallin	Symétrie Caractéristique	Exigences	Intuitions
Isométrique	4 axes ternaires de rotation	$a = b = c, \alpha = \beta = \gamma = 90^\circ$	Symétrie maximale : cubes, octaèdres.

Système Cristallin	Symétrie Caractéristique	Exigences	Intuitions
Tétragonal	1 axe quaternaire de rotation	$a = b, \alpha = \beta = \gamma = 90^\circ$	Comme le cubique, mais avec un axe allongé (c).
Hexagonal	1 axe sennaire de rotation	$a = b, \alpha = \beta = 90^\circ, \gamma = 120^\circ$	Symétrie élevée avec un axe principal (ex. : prismes hexagonaux).
Trigonal	1 axe ternaire de rotation	$a = b, \alpha = \beta = 90^\circ, \gamma = 120^\circ$	Similaire à l'hexagonal, mais avec une symétrie ternaire (ex. : quartz).
Orthorhombique	3 axes binaires de rotation	$\alpha = \beta = \gamma = 90^\circ$	Étendre la logique monoclinique à 3 axes (ex. : prismes orthorhombiques).
Monoclinique	1 axe binaire de rotation	$\alpha = \gamma = 90^\circ$	Rotation de 180° autour de l'axe c, sans créer de vides dans le plan de base.
Triclinique	Aucune, ou seulement un centre de symétrie	aucune	Symétrie minimale : pas de rotation, seulement une inversion possible.

Classes Cristallines, Symétries et Formes

Système Cristallin	Classe Cristalline	Hermann-Mauguin	Symétries	Formes Courantes	Groupe Abstrait
Isométrique	Hexoctaédrique	4/m 3 2/m	3 A_4 , 4 A_3 , 6 A_2 , 9 M, C	Cube {100}, octaèdre {111}, dodécaèdre {110}	$S_4 \times Z_2$
Isométrique	Diploïdale	2/m -3	3 A_2 , 4 A_3 , C	Cube {100}, diploïde {321}	$A_4 \times Z_2$
Tetragonale	Ditetragonale dipyramidale	4/m 2/m 2/m	1 A_4 , 4 A_2 , 5 P, C	Prismes, dipyramides	$D_8 \times Z_2$
Hexagonal	Dihexagonale dipyramidale	6/m 2/m m	1 A_6 , 6 A_2 , 6 M, C	Prisme hexagonal {10-10}, bipyramide hexagonale {11-21}	$D_{12} \times Z_2$
Trigonal	Rhomboédrique	-3m	1 A_3 , 3 A_2 , 3 M, C	Rhomboèdre {10-11}, scalénoèdre {21-31}	Cyclique $Z_6 = Z_2 \times Z_3$
Trigonal	Trapézoédrique	3 2	1 A_3 , 3 A_2	Prisme trigonal {10-10}, rhomboèdre {10-11}	diédrale $D_6 = D_2 \times D_3$

Système Cristallin	Classe Cristalline	Hermann-Mauguin	Symétries	Formes Courantes	Groupe Abstrait
Orthorhombique	Dipyramide orthorhombique	2/m 2/m 2/m	3 A ₂ , 3 P, C	Pinacoïdes : {100}, {010}, {001} ; prismes : {100}, {010}, {001}, dipyramide {111}	Klein 4 V = Z ₂ x Z ₂
Monoclinique	Prismatique	2/m	1 A ₂ , 1 P, C	Pinacoïdes : {100}, {010}, {001} ; prismes : {0kl}, {hk0}	Klein 4 V = Z ₂ x Z ₂
Triclinique	Pinacoïdale	1	C	Pinacoïdes : {h00}, {0k0}, {00l}, {h0l}, {0kl}, {hk0}, {hkl} (2 faces chacun)	Cyclique Z ₂

20 Minéraux de Référence

Minéral	Système Cristallin	Classe Cristalline	Hermann-Mauguin	Formes Typiques	Groupe d'Espace	Formule Chimique
Diamant	Isométrique	Hexoctaédrique	m-3m	Octaèdre {111}, cube {100}, dodécaèdre {110}	Fd-3m	C
Fluorite	Isométrique	Hexoctaédrique	m-3m	Cube {100}, octaèdre {111}, dodécaèdre {110}	Fm-3m	CaF ₂
Pyrite	Isométrique	Diploïdale	m-3	Cube {100}, diploïde {321}	Pa-3	FeS ₂
Almandin (Grenat)	Isométrique	Hexoctaédrique	m-3m	Dodécaèdre rhomboïdal {110}	Ia-3d	Fe ₃ Al ₂ (SiO ₄) ₃
Halite	Isométrique	Hexoctaédrique	m-3m	Cube {100}	Fm-3m	NaCl
Leucite	Isométrique	Hexoctaédrique	m-3m	Trapézoèdres	Ia-3d	KAlSi ₂ O ₆

Minéral	Système Cristallin	Classe Cristalline	Hermann-Mauguin	Formes Typiques	Groupe d'Espace	Formule Chimique
Grenat (Pyrope)	Isométrique	Hexoctaédrique	m-3m	Dodécaèdre rhomboïdal {110}	Ia-3d	$\text{Mg}_3\text{Al}_2(\text{SiO}_4)_3$
Béryl	Hexagonal	Hexagonale	6/mmm	Prismes hexagonaux allongés	P6/mmm	$\text{Be}_3\text{Al}_2(\text{SiO}_3)_6$
Graphite	Hexagonal	Hexagonale	6/mmm	Feuillets hexagonaux	P6 ₃ /mmc	C
Apatite	Hexagonal	Hexagonale	6/m	Prismes hexagonaux, aiguilles	P6 ₃ /m	$\text{Ca}_5(\text{PO}_4)_3(\text{OH}, \text{F}, \text{Cl})$
Quartz	Trigonal	Trapézoédrique	32	Prismes hexagonaux, bipyramides trigonales	P3 ₁ 21	SiO_2
Calcite	Trigonal	Rhomboédrique	-3m	Rhomboèdre {10-11}, scalénoèdre {21-31}	R-3c	CaCO_3
Corindon	Trigonal	Rhomboédrique	-3m	Dodécaèdre rhomboïdal, barils	R-3c	Al_2O_3
Vésuvianite	Tétragonal	Bipyramidale tétragonale	4/mmm	Prismes courts, pyramides tétraédriques	P4/nnc	$\text{Ca}_{10}(\text{Mg}, \text{Fe})_2\text{Al}_4(\text{SiO}_4)_5(\text{Si}_2\text{O}_7)_2(\text{OH})_4$
Topaze	Orthorhombique	Prismatique	mmm	Prismes, pyramides	Pnma	$\text{Al}_2\text{SiO}_4(\text{F}, \text{OH})_2$
Aragonite	Orthorhombique	Rhomboédrique	mmm	Prismes, aiguilles	Pmcn	CaCO_3
Orthose	Monoclinique	Prismatique	2/m	Tablettes, macles de Karlsbad	C2/m	KAlSi_3O_8
Gypse	Monoclinique	Prismatique	2/m	Prismes, masses fibreuses, tablettes	P2 ₁ /c	$\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$
Heulandite	Monoclinique	Prismatique	2/m	Tablettes, prismes	C2/c	$(\text{Na}, \text{Ca})_4\text{Al}_6\text{Si}_{10}\text{O}_{32} \cdot 12\text{H}_2\text{O}$
Talc	Triclinique	Pinacoïdale	1	Feuillets, masses micacées	P-1	$\text{Mg}_3\text{Si}_4\text{O}_{10}(\text{OH})_2$